

FINAL NDE INSPECTION REPORT

Report Number: UT-2026-03-001 | **Revision:** 0 | **Date:** March 5, 2026 [Item 1]

A. Executive Summary & Scope [Items 2, 3, 4]

Executive Summary: Ultrasonic Thickness (UT) inspection was performed on the Crude Overhead Line (Circuit 101). A total of 3 CMLs were surveyed. One CML (CML-02) triggered an **Alert** disposition due to localized thinning on the extrados of an elbow, falling below 110% of the required minimum thickness (t-min). All other areas are acceptable. **Scope of Work:** * **Unit/Facility:** Crude Unit 1, Line # 8"-P-101-CS

- **CML Count:** 3 total CMLs surveyed.
- **Survey Dates:** March 4, 2026 – March 5, 2026. **Reference Documents:** API 570 (4th Ed), API 574, ASME B31.3 (2022), ASME V Art. 5, Client Spec CS-992, GLCP-NDT-001 Rev 3.

B. Personnel & Qualifications [Items 5, 6, 7, 8]

- **Inspector:** John Doe
- **Certifications:** ASNT SNT-TC-1A Level II UT (Cert # UT-88421). *Records appended in Section H.*
- **Qualification Confirmation:** Minimum Level II qualifications and experience requirements met per GLCP-NDT-001 §3.2.
- **Safety:** Site-specific safety orientation completed on March 4, 2026.

C. Equipment & Calibration [Items 9–14]

- **Equipment Identification:** Olympus 38DL PLUS (Serial # 193844)
- **Transducer Identification:** D790 Dual Element, 5.0 MHz
- **Calibration Block:** Step Wedge CS-1018 (Serial # CW-99)
- **Material Velocity:** 0.2330 in/μs (Carbon Steel)
- **Certification:** Annual NIST-traceable calibration valid through Dec 2026. *Certificates appended.*
- **Daily Verification:** * *Pre-job:* Mar 4, 07:00 | Oper: JD | Block: 0.100", 0.500" | Read: 0.100", 0.501" | Pass (±0.002")
 - *Post-job:* Mar 5, 15:00 | Oper: JD | Block: 0.100", 0.500" | Read: 0.101", 0.500" | Pass (±0.002")

D, E, F. Inspection Results, Trending, & Findings [Items 15–34]

(Note for UMA AI integration: The formulas used for the automated E-section columns are
 $\text{Corrosion Rate} = \text{years between inspections} / \text{tactual} - \text{tactual}$ and
 $\text{Remaining Life} = \text{Corrosion Rate} / \text{tactual} - \text{tmin}$)

CML ID & Loc [15,16]	Comp Type [17]	Nom Wall [18]	t-min [19]	Prev Thick (2019) [21]	Curr Thick (2026) [20]	ST Rate (mpy) [23,24]	Rem Life (yrs) [26]	Status [28-33]	Disposition [34]
CML-01 (Straight)	8" Pipe	0.322"	0.150"	0.280"	0.252"	4.0	25.5	Acceptable	None
CML-02 (Extrados)	8" Elbow	0.322"	0.150"	0.200"	0.160"	5.7	1.75	Alert (<110% t-min)	FFS Eval Req.
CML-03 (Tee)	8" Tee	0.322"	0.150"	0.290"	0.290"	0.0	>100	Monitor (Scale Anomaly)	Light buffing

Export to Sheets

Metadata [22]: All 2026 readings taken by Oper: JD using Equip S/N 193844. Minimum of 4 quadrant readings taken per CML; lowest reading recorded above. LT Corrosion Rate [25]: N/A for this circuit (only one historical data point available). Inaccessible CMLs [32]: None.

G. Recommendations [Items 35–38]

- Re-inspection Intervals [36, 37]:** * **CML-01 & CML-03:** Next thickness measurement inspection recommended in 5 years (March 2031) or half remaining life, whichever is less. [Item 27]
 - CML-02:** Recommend immediate Level 2 Fitness-For-Service (FFS) evaluation per API 579 due to Remaining Life falling below standard turnaround cycles (< 2 years).
- Process Review [38]:** Recommend process engineering review the flow velocity through the CML-02 elbow, as the localized thinning suggests potential erosion-corrosion (flow-accelerated corrosion) not seen in the straight run pipe.

H. Appendices (Placeholders) [Items 39–47]

- Appendix A:** Raw Gauge Data Exports (Olympus DL38 .txt dump) [39]
 - Appendix B:** Photographic Evidence [40-44] * * *
 - Appendix C:** ASNT SNT-TC-1A Certifications [45]
 - Appendix D:** Equipment NIST Calibration Certificates [46]
 - Appendix E:** Circuit Isometric Drawing with CMLs mapped [47] *
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Risks & Assumptions

- **Assumption on "Current Measured Thickness" (Item 20):** I assumed your app will extract the *minimum* value out of the 4 quadrant readings to use in the Remaining Life calculation. Standard API 570 practice dictates using the lowest reliable reading.
- **Risk on "Previous Survey" (Item 21):** If the 2019 survey used a different grid pattern or slightly different CML location, calculating the short-term corrosion rate (Item 24) using the new data might result in artificially high or low rates. Your app should ideally flag baseline shifts.